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State Transition Diagram Testing

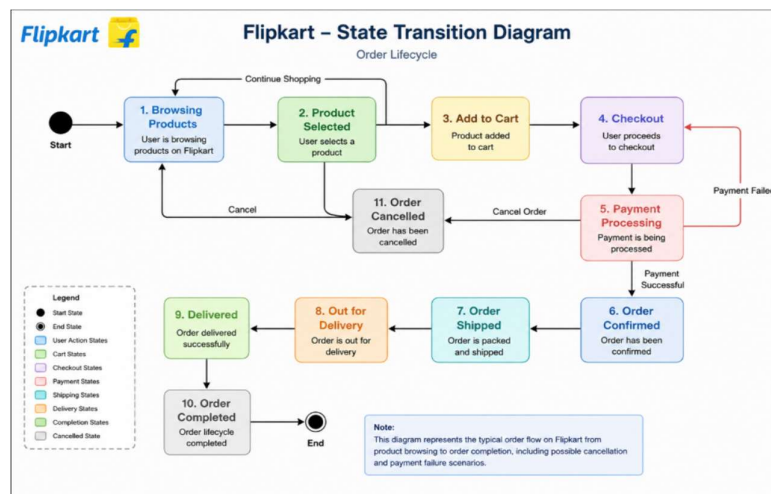
Introduction

State Transition Testing is a black-box testing technique used to verify that a system behaves correctly as it moves from one state to another based on user actions or events. It ensures that every valid and invalid transition is tested and that the application follows the expected workflow.

Objective

- To verify that the application moves correctly from one state to another.
- To ensure invalid transitions are handled properly.
- To validate the complete user workflow.
- To improve the reliability and consistency of the application.

1. Flipkart – State Transition Diagram



States

- Browse Products
- Product Selected
- Add to Cart
- Checkout
- Payment Processing
- Order Confirmed
- Order Shipped

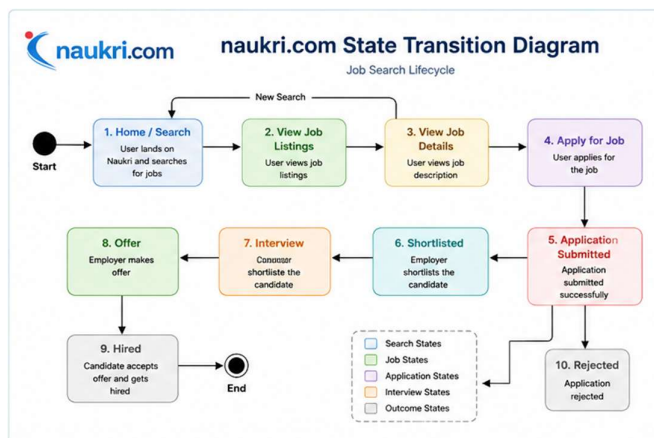
- Out for Delivery
- Delivered
- Order Completed
- Order Cancelled

TC ID	Current State	Event	Expected Next State
ST1_01	Browse Products	Select Product	Product Selected
ST1_02	Product Selected	Add to Cart	Add to Cart
ST1_03	Checkout	Payment Successful	Order Confirmed
ST1_04	Checkout	Payment Failed	Payment Failed
ST1_05	Order Confirmed	Ship Order	Order Shipped
ST1_06	Order Shipped	Deliver Order	Out for Delivery
ST1_07	Out for Delivery	Delivery Successful	Delivered
ST1_08	Payment Processing	Cancel Order	Order Cancelled

Explanation

These test cases verify that the Flipkart application moves correctly between different order states and properly handles exceptional cases such as payment failure and order cancellation.

2. Naukri – State Transition Diagram



States

- Home/Search
- View Job Listings
- View Job Details

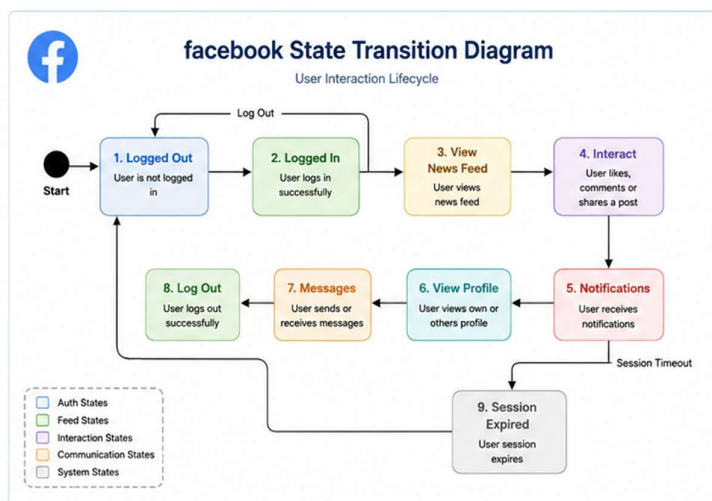
- Apply for Job
- Application Submitted
- Shortlisted
- Interview
- Offer
- Hired
- Rejected

TC ID	Current State	Event	Expected Next State
ST2_01	Home	Search Job	View Job Listings
ST2_02	Job Listing	Select Job	View Job Details
ST2_03	Job Details	Apply	Application Submitted
ST2_04	Submitted	Employer Shortlists	Shortlisted
ST2_05	Shortlisted	Interview Completed	Offer
ST2_06	Submitted	Employer Rejects	Rejected

Explanation

These test cases validate that the job application process follows the correct sequence and reaches the appropriate outcome.

3. Facebook – State Transition Diagram



States

- Logged Out
- Logged In

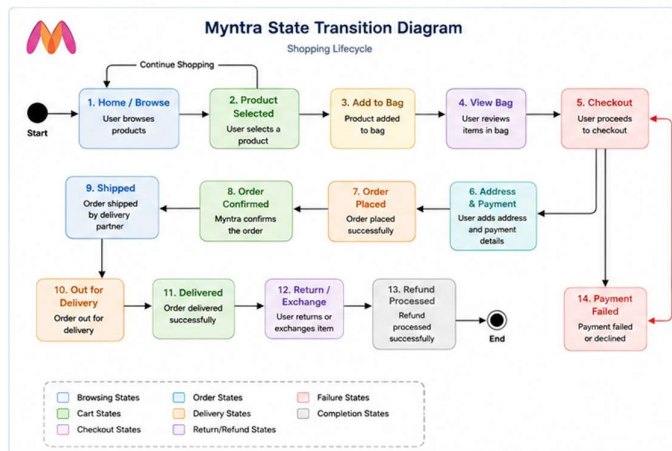
- View News Feed
- Interact
- Notifications
- View Profile
- Messages
- Log Out
- Session Expired

TC ID	Current State	Event	Expected Next State
ST3_01	Logged Out	Login	Logged In
ST3_02	Logged In	Open Feed	View News Feed
ST3_03	News Feed	Like/Comment	Interact
ST3_04	Interact	Receive Notification	Notifications
ST3_05	Logged In	Logout	Logged Out
ST3_06	Logged In	Session Timeout	Session Expired

Explanation

These test cases verify successful user navigation and proper session handling.

4. Myntra – State Transition Diagram



States

- Browse Products
- Product Selected
- Add to Bag
- View Bag

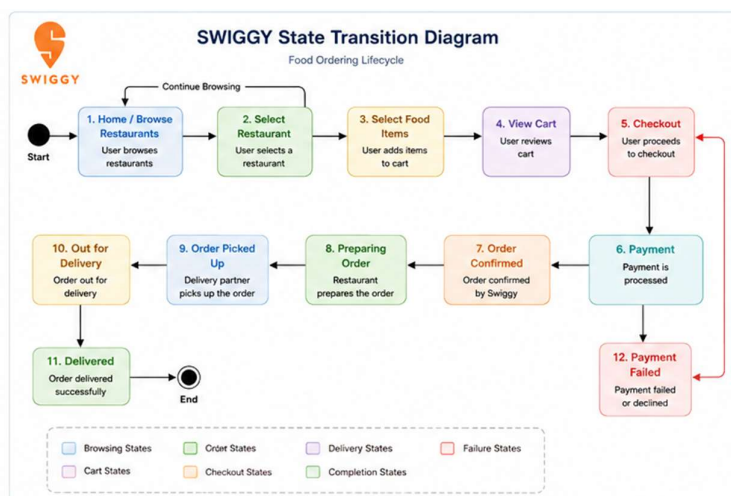
- Checkout
- Payment
- Order Placed
- Order Confirmed
- Shipped
- Out for Delivery
- Delivered
- Return/Exchange
- Refund Processed
- Payment Failed

TC ID	Current State	Event	Expected Next State
ST4_01	Browse	Select Product	Product Selected
ST4_02	Product Selected	Add to Bag	Add to Bag
ST4_03	Checkout	Payment Success	Order Placed
ST4_04	Checkout	Payment Failure	Payment Failed
ST4_05	Delivered	Return Request	Return/Exchange
ST4_06	Return Approved	Refund	Refund Processed

Explanation

These test cases validate both successful shopping and return/refund workflows.

5. Swiggy – State Transition Diagram



TC ID	Current State	Event	Expected Next State
ST5_01	Browse Restaurants	Select Restaurant	Restaurant Selected
ST5_02	Select Food	Add to Cart	View Cart
ST5_03	Checkout	Payment Success	Order Confirmed
ST5_04	Checkout	Payment Failure	Payment Failed
ST5_05	Preparing Order	Picked Up	Order Picked Up
ST5_06	Out for Delivery	Delivered	Delivered

Explanation

These test cases ensure the application correctly transitions through each stage of the food delivery process and handles payment failures appropriately.